

Designing ELECTRO-MEDICAL EQUIPMENT

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An overview of the technology upgrades and essentials in medical electronics design

Designing electronic medical devices requires engineers to walk an extra mile. After all, it's the matter of saving lives! Since accuracy and reliability are crucial for medical equipment, designers have to follow a strict set of regulations for their designs. While this norm has been a constant in this vertical, there have been many changes and upgradations in terms of form factor, usage and technology incorporated in devices. This article reviews all these upgrades and essentials in medical electronics design.

New technologies

Medical devices—both diagnostic and treatment types—have seen a shift in design preferences over the years. The combination of electronics, mechanical engineering, medical science and software engineering governs the progress of these equipment.

Nitesh K. Jangir, co-founder, Coeo Labs, a Bengaluru-based medical equipment manufacturer, says, "Design starts from the hardware. Most of the medical devices use embedded software. So hardware requires more attention. You need to know what kind of sensors, components and

MCUs you will be using."

For medical devices, application-specific integrated circuit (ASIC) system-on-chips are preferable over field-programmable gate array (FPGA) chips—the reason being that although FPGA offers more customisation options through simple designs, ASICs provide higher power efficiency, better performance and purpose-specificity. Upgraded image processing technologies have improved different applications such as computed tomography and magnetic resonance imaging scan tests.

Miniaturisation and portability is another major trend in medical devices. To ensure easy handling as well as quick diagnosis, many critical equipment like ultrasound machines and electrocardiograms (ECG) are being designed in portable shape. This has made diagnosis of commute-affected patients much easier.

Even ventilators are becoming portable. One of the newest examples is a portable ventilator designed by professor Diwakar Vaish, head of robotics and research, A-SET Training and Research Institute, in collaboration with Dr Deepak Agarwal, professor of neurosurgery at AIIMS, New Delhi. Its main components include pressure sensor monitors and a special microcontroller unit that operates on a program based on machine learning algorithm.

As regards connectivity, Nitesh says, "Almost all medical devices are becoming connected to the Internet. This is giving users the freedom to utilise data from anywhere. For instance, users are adopting ECG devices with Bluetooth connectivity to access ECG data through their phones. We as designers try to ensure that every device at least has the option to connect to the Internet. Wi-Fi, GSM and Bluetooth are the main wireless technologies being utilised. So we integrate the respective modules with the circuit board."

"Globally, Zigbee is widely used in medical equipment. In India, there are a handful of companies using Zigbee. These companies use Zigbee for medical equip-

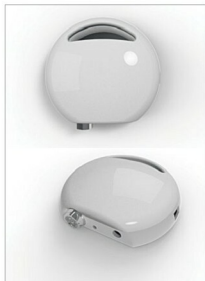


Fig. 1: Portable ventilator by A-SET and AIIMS

FPGAs Vs ASICs

FPGAs	ASIC-SoCs
Faster time to market—no layout, masks and manufacturing steps needed	Need longer design times to take care of all manufacturing steps
Field reprogrammability—design changes can be absorbed even in field and FPGA reprogrammed	Once manufactured, need to spin again a new chip in case of bugs
More power consumption and may not be high in performance because of programmable design and low clock speeds	Custom design for an application helps in designing for power/performance efficiencies
Good for prototyping and low-volume designs as cost would be less	For larger volume of production, cost per unit will be much less for an ASIC
Generally, not possible to have analogue/mixed-signal designs and limited to what vendor supports	Can support analogue and mixed-signal designs